

Finance Atlas

A complete map of the finance curriculum we agreed to build.

Explore the topics, follow their relationships, and turn curiosity into a learning journey.

40

finance areas

678

original topic units

698

concept nodes

2520

labelled connections

The workbook contains the full network. This PDF provides the overview, precise connection diagrams, all 40 topic inventories and 24 proposed learning journeys.

The 678 original bullets remain intact. Twenty atomic anchors make important distinctions explicit. This is a content plan: full lessons and final quiz items have not yet been authored.

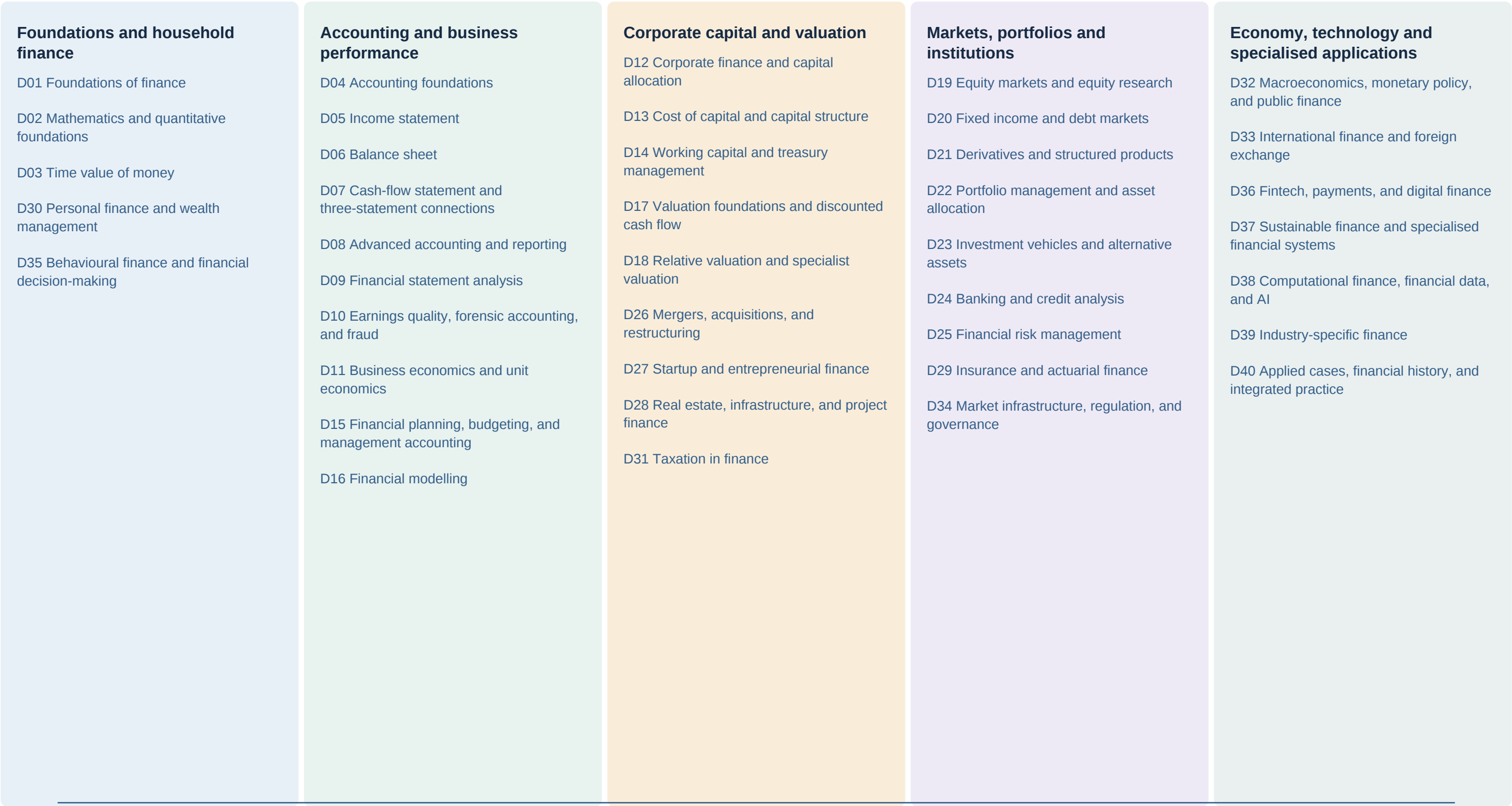
How to read and use this atlas

- | | |
|---|--|
| 1. Choose an entry point | Start from a statement, a company, a question or a curriculum area. The same stable concept IDs connect all four routes. |
| 2. Distinguish the link types | PART_OF is classification. RECOMMENDED_BEFORE is optional preparation. EXPLORE_WITH is a topical companion. Financial links explain mechanisms or applications. |
| 3. Follow the local web | Each PDF diagram shows a small part of the network. In the workbook, filter Connections by From ID or To ID to see all connections for a concept. |
| 4. Preserve context while going deeper | A concept should open alongside the statement or problem. More depth reveals calculations, examples, assumptions and connected topics. A trail brings you back. |
| 5. Use a question to move forward | Predict first, inspect the evidence, explain the result, and follow the next connection. The named journeys are proposed sequences, not rigid gates. |
| 6. Separate mapping from completed content | Every original topic is mapped. Opening questions, stages and interactions are authoring proposals. Source-specific definitions and company examples are later work. |

Navigation: linked IDs jump to topic locations in this PDF. Viewer support can vary. The same IDs can always be searched in the workbook.

The complete finance landscape

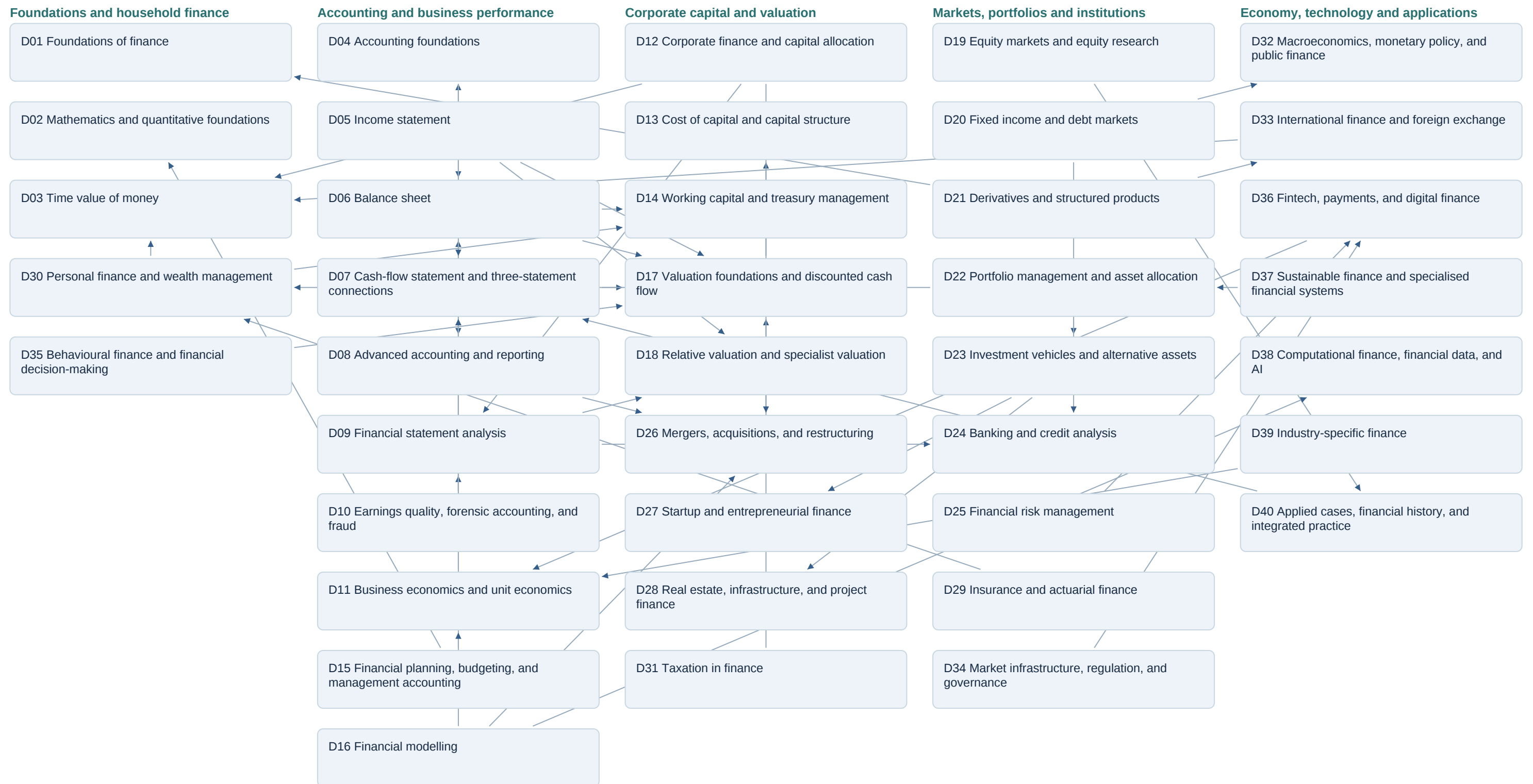
Five viewing groups organise the same network. Follow the detailed diagrams and concept IDs to explore actual relationships.



Foundations prepare the tools. Statements explain business performance. Corporate finance allocates and values capital. Markets and institutions price and channel it. Economic, technological and sector contexts connect across the whole map.

One connected web across all 40 finance areas

45 selected financial and application bridges connect every area. The next two pages explain the arrows using exact topic IDs.



Read an arrow as a connection between particular concepts in the two areas, not as a claim that one entire area causes the other. Membership and preparation links are excluded from this overview. The workbook contains all 2,520 relationships.

What the cross-area arrows mean

Selected examples from the complete relationship table. IDs link to the corresponding topic locations.

[D06 > D14 A02 > A05](#)

component of: Operating receivables are one component of operating working capital.

[D06 > D17 T06.02 > T17.15](#)

assessed in: Cash is assessed for operating need before it is treated as excess cash.

[D06 > D08 T06.09 > T08.06](#)

reported under: Right-of-use assets arise from applicable lease accounting.

[D39 > D11 T39.07 > T11.10](#)

uses sector model: Marketplace analysis requires customer, transaction and take-rate economics.

[D30 > D03 T30.05 > T03.10](#)

uses: Loan comparisons require payment, amortisation and cost calculations.

[D05 > D18 T05.18 > T18.04](#)

denominator for: Earnings per share is the denominator of a share-price P/E ratio.

[D10 > D06 T10.05 > T06.17](#)

tests: Provision and reserve movements can affect reported earnings.

[D04 > D06 T04.01 > T06.18](#)

explains: The accounting equation makes funding claims reconcile with the asset side.

[D07 > D17 A07 > A11](#)

deducted in: Cash operating capital expenditure is deducted when calculating standard FCFF.

[D11 > D09 T11.05 > T09.11](#)

explains: Fixed operating costs and utilisation contribute to operating leverage.

[D31 > D17 T31.07 > A10](#)

affects: Tax depreciation affects operating cash taxes separately from book depreciation.

[D19 > D40 T19.12 > T40.18](#)

tested by: Investment theses should be compared with subsequent evidence and outcomes.

[D20 > D32 T20.13 > T32.06](#)

responds to: Floating coupons reset against specified reference-rate conventions.

[D22 > D23 T22.10 > T23.03](#)

implemented through: Active and passive choices can be implemented through suitable funds.

[D23 > D27 T23.09 > T27.05](#)

provides capital: Venture and growth investors finance startup funding rounds.

[D25 > D36 T25.13 > T36.11](#)

managed with: Identity and fraud controls address selected digital operational risks.

[D34 > D36 T34.10 > T36.11](#)

supported by: Financial identity checks and monitoring support anti-fraud and AML controls.

[D12 > D03 T12.05 > T03.12](#)

calculation input: Incremental project cash flows are discounted to calculate project NPV.

[D33 > D03 T33.03 > T03.09](#)

relates to: Purchasing-power comparisons relate price levels and exchange rates.

[D05 > D17 A08 > A11](#)

added back in: Non-cash depreciation deducted in operating profit is added back in the standard FCFF build.

[D16 > D07 T16.03 > T07.14](#)

models: A three-statement model connects transaction effects across the statements.

[D40 > D07 T40.02 > T07.14](#)

practises: Trace the same transaction through all three statements.

[D08 > D26 T08.01 > T26.11](#)

applied in: Acquisition accounting feeds the pro-forma merger model.

What the cross-area arrows mean

Selected examples from the complete relationship table. IDs link to the corresponding topic locations.

[D12 > D09 T12.15 > A20](#)

assessed with: Capital allocation can be assessed through operating returns and value created.

[D09 > D18 T09.16 > T18.07](#)

supports: Normalisation supports comparable valuation denominators.

[D09 > D24 T09.07 > T24.14](#)

informs: Leverage and coverage metrics inform repayment capacity and covenant analysis.

[D15 > D11 T15.08 > T11.10](#)

informs: Costing methods affect measured product and customer profitability.

[D36 > D11 T36.02 > T11.01](#)

has economics: Card payments distribute fees and responsibilities across participants.

[D17 > D13 A11 > A13](#)

discounted at: FCFF is discounted at a matching WACC in a standard enterprise DCF.

[D16 > D38 T16.16 > T38.18](#)

supported by: Reproducibility and checks support model auditability.

[D35 > D17 T35.03 > T17.18](#)

can bias: Anchoring can distort valuation assumptions and scenario interpretation.

[D20 > D24 T20.12 > T24.11](#)

reflects partly: Credit spreads reflect default risk along with liquidity and other premiums.

[D21 > D33 T21.01 > T33.08](#)

hedges: Currency forwards and futures can hedge specified currency exposures.

[D22 > D30 T22.03 > T30.10](#)

informs: Risk capacity and horizon affect retirement investment choices.

[D37 > D22 T37.01 > T22.02](#)

informs: ESG objectives and risks can be reflected in investment policy and analysis.

[D23 > D28 T23.12 > T28.04](#)

values exposure: Real estate funds and REITs depend on property economics and valuation.

[D29 > D30 T29.15 > T30.10](#)

supports: Annuities and pensions can provide retirement income.

[D21 > D01 T21.05 > T01.15](#)

derived from: Put-call parity is an arbitrage relationship under its assumptions.

[D15 > D02 T15.12 > T02.14](#)

uses: Capacity and product-mix decisions can use constrained optimisation.

[D10 > D04 T10.02 > T04.06](#)

tests: Revenue-quality analysis tests whether recognition reflects applicable criteria.

[D04 > D07 T04.02 > T07.14](#)

enables: Double entry lets one transaction be traced through accounts and statements.

[D10 > D07 T10.06 > T07.03](#)

tests: Accrual quality is assessed partly by reconciling earnings to operating cash.

[D12 > D26 T12.14 > T26.01](#)

compares with: Acquisitions and organic investment are alternative routes for deploying capital.

[D30 > D14 T30.02 > T14.04](#)

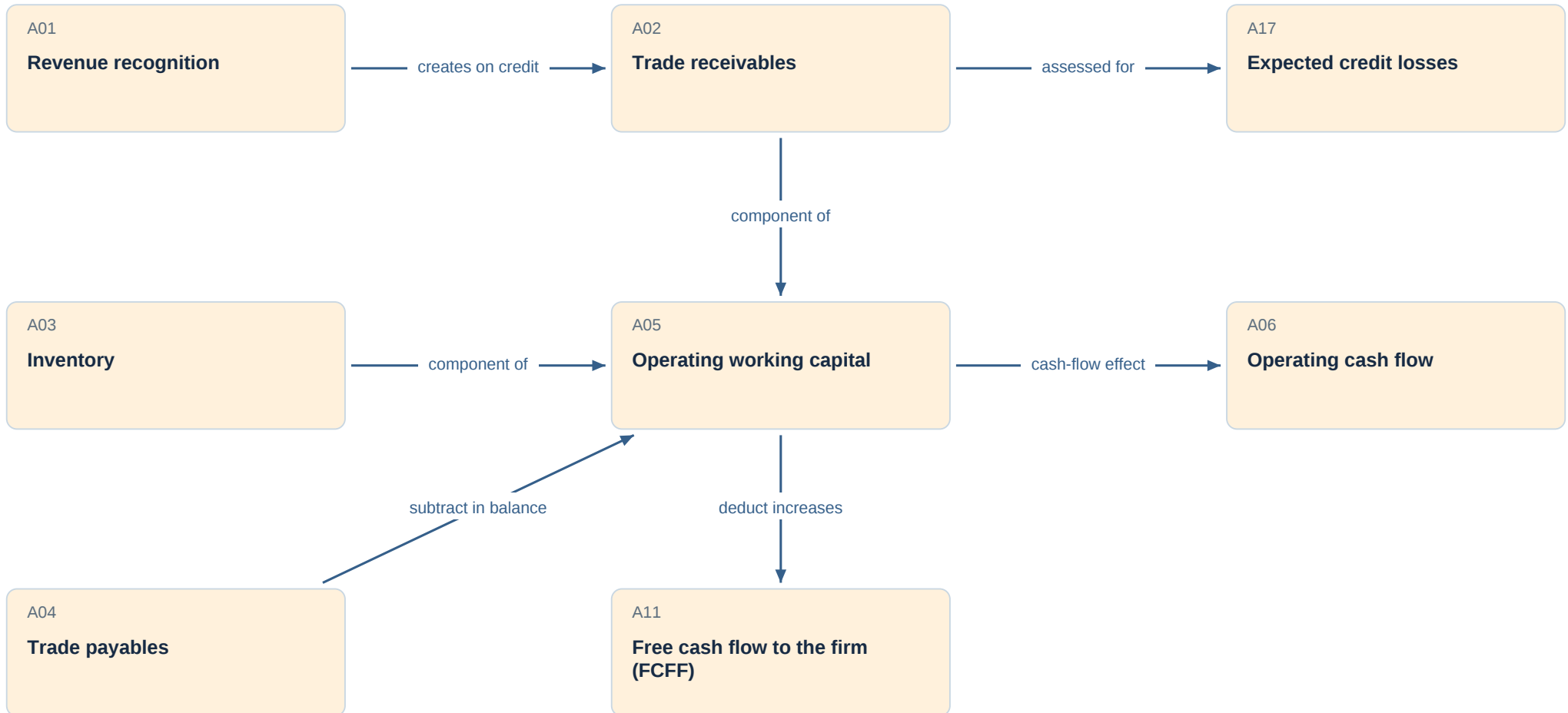
analogous to: Household budgeting and business cash budgeting both track receipts and payments.

[D16 > D26 T16.12 > T26.11](#)

applied in: Merger modelling combines standalone forecasts, terms, financing and adjustments.

From sales to working capital and cash

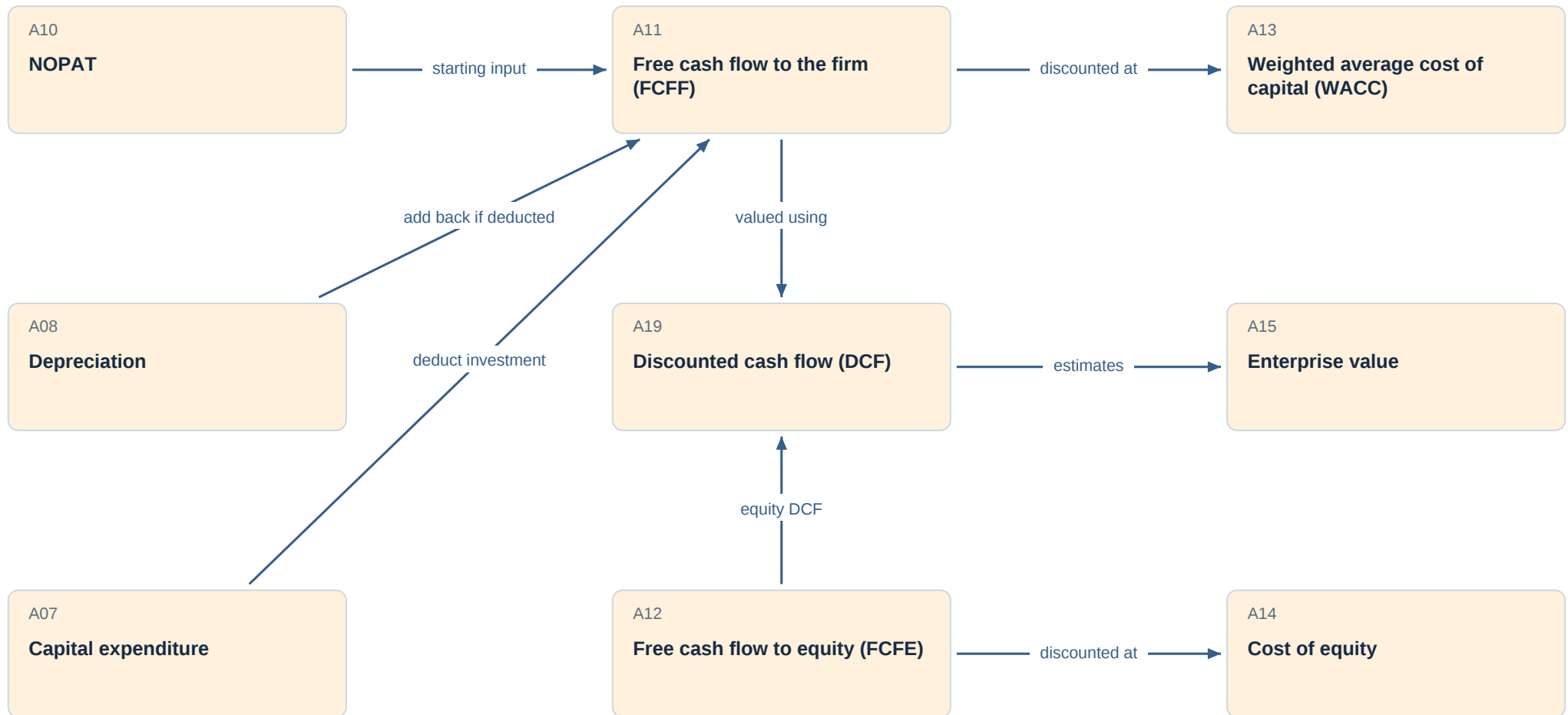
Why can a sale increase profit before cash arrives?



Operating working capital uses an explicit operating definition. A rise generally uses cash, all else equal. Credit-loss recognition and cash-flow classification depend on the applicable reporting framework.

Free cash flow, discount rates and value

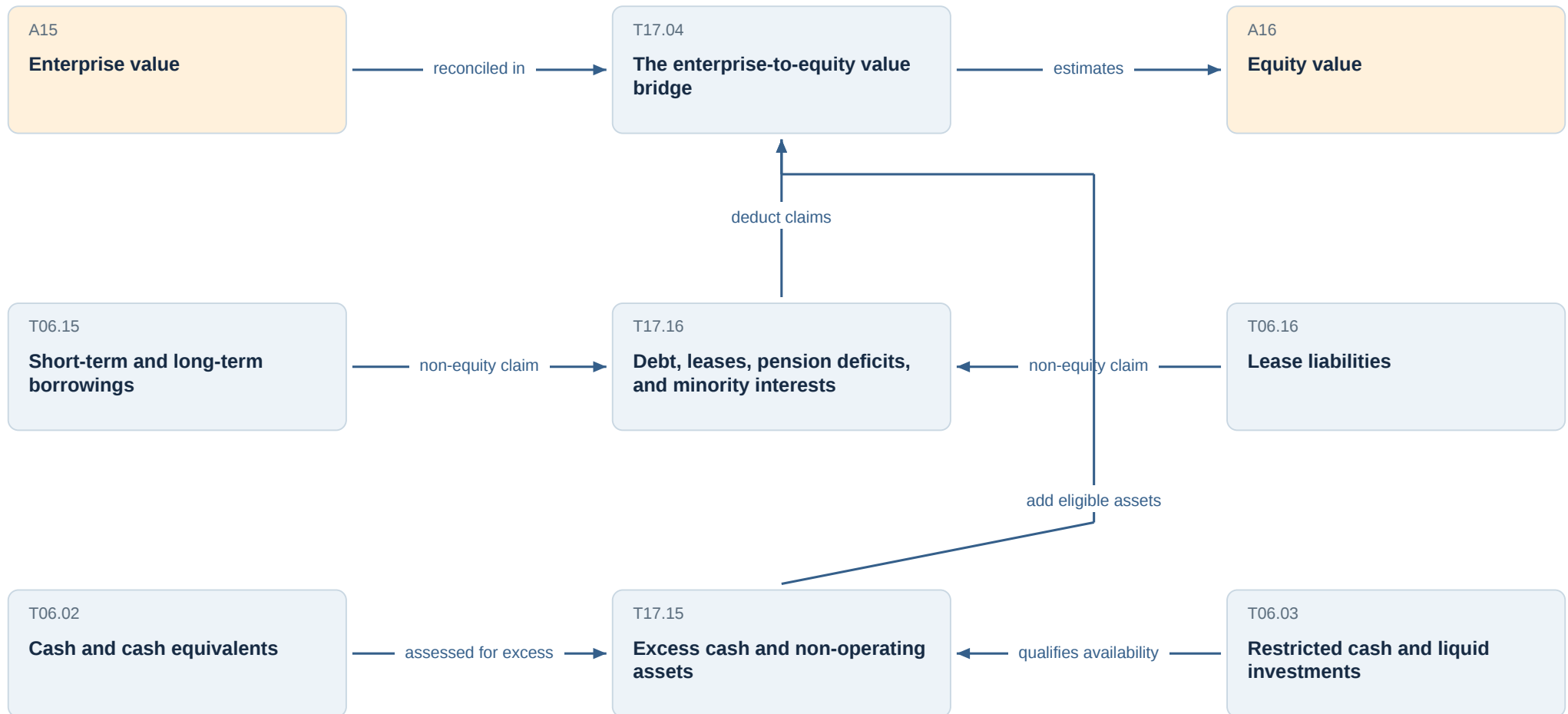
Which cash belongs to all investors and which belongs to equity?



Standard FCFF: NOPAT + applicable non-cash charges - operating capex - increase in operating working capital. FCFE includes the relevant debt cash flows. Match each cash flow to its discount rate and investor claim.

Assets, funding claims and equity value

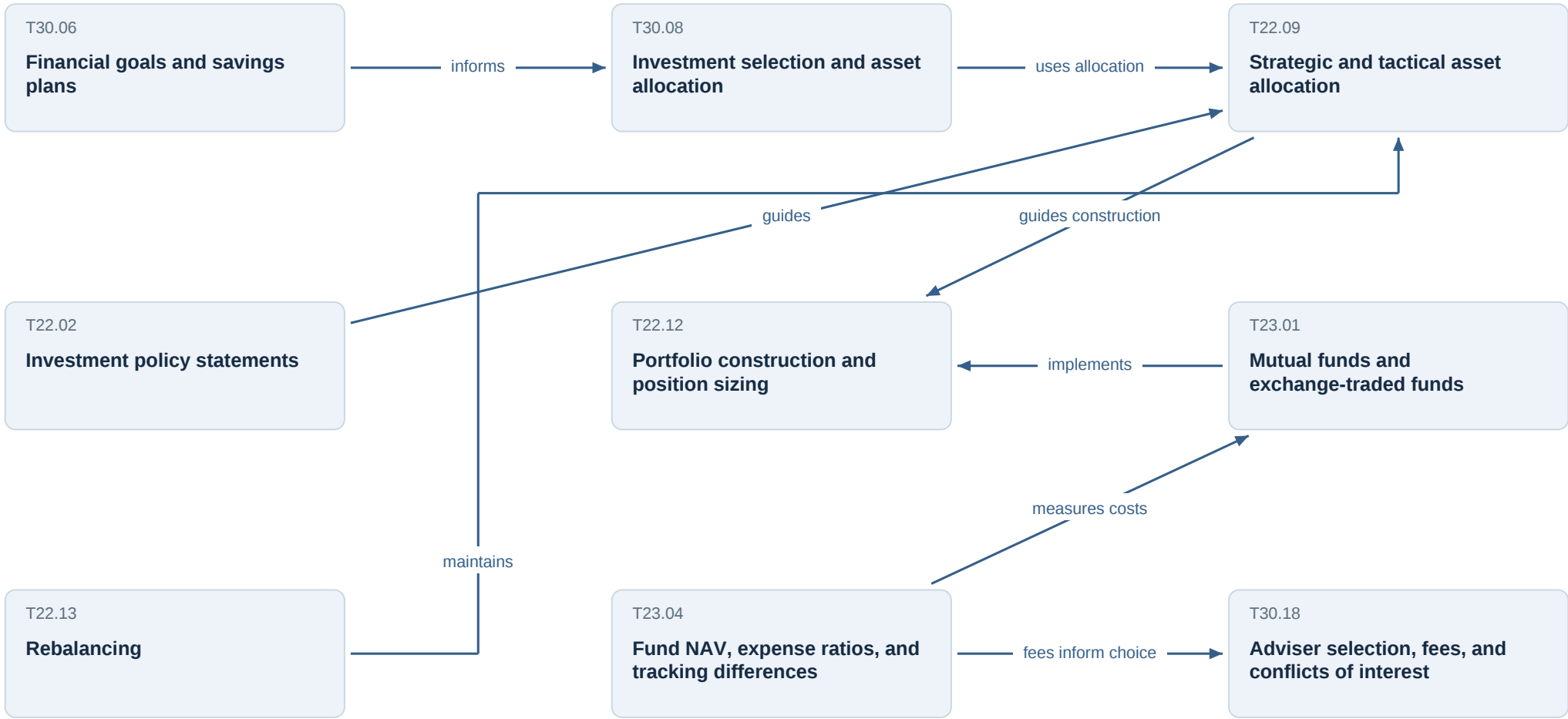
Why does the enterprise-to-equity bridge need careful definitions?



Add relevant non-operating assets and deduct non-equity claims under a consistent valuation perimeter. Do not add all reported cash automatically or subtract claims already reflected in the cash-flow treatment.

Goals, diversification and investment implementation

How do household goals connect to portfolio choices?



The goal defines the decision context; portfolio choices also need constraints, risk capacity and liquidity needs. Funds implement exposure, while fees and structure affect investor outcomes. View all remaining links by ID in the workbook.

Precise anchors inside the broader topics

These 20 nodes enable separate click targets without dropping the original grouped topic units.

A01	Revenue recognition	Revenue is recognised when the applicable recognition criteria are met, which can differ from cash collection.
A02	Trade receivables	Amounts owed by customers for recognised sales; collection converts the receivable into cash.
A03	Inventory	Goods held for sale or used in production, measured under the applicable accounting framework.
A04	Trade payables	Amounts owed to suppliers for goods or services already received.
A05	Operating working capital	Operating current assets less operating current liabilities, using an explicit definition that usually excludes cash and financing debt.
A06	Operating cash flow	Cash generated or used by operating activities, classified under the applicable reporting framework.
A07	Capital expenditure	Cash invested in qualifying long-lived operating assets; it is distinct from the period depreciation expense.
A08	Depreciation	Allocation of a depreciable asset cost over its useful life; generally a non-cash expense in the period.
A09	EBIT	Earnings before interest and tax; specify adjustments and reconcile it with reported operating profit.
A10	NOPAT	After-tax operating profit for the specified operating perimeter; a simplified model uses EBIT multiplied by one minus an operating tax rate.

Precise anchors inside the broader topics

These 20 nodes enable separate click targets without dropping the original grouped topic units.

A11	Free cash flow to the firm (FCFF)	Cash available to all capital providers after operating taxes and reinvestment, before financing distributions.
A12	Free cash flow to equity (FCFE)	Cash available to equity after reinvestment, debt cash flows and other relevant financing claims.
A13	Weighted average cost of capital (WACC)	Market-value-weighted required return on the financing mix, with applicable debt tax treatment.
A14	Cost of equity	Required return of equity investors for the risk of the equity claim.
A15	Enterprise value	Value of the specified operating enterprise available to its capital providers; reconcile the perimeter to equity value.
A16	Equity value	Value attributable to common equity after the relevant non-equity claims and non-operating adjustments.
A17	Expected credit losses	Probability-weighted credit-loss estimates under the applicable impairment framework.
A18	Net borrowing	Cash proceeds from new debt less cash principal repayments over the specified period.
A19	Discounted cash flow (DCF)	Value expected future cash flows by discounting at a rate matched to their risk, currency and investor claim.
A20	Return on invested capital (ROIC)	After-tax operating profit relative to the invested capital used to generate it, using consistent timing and scope.

Question-led journeys through the web

Each route reuses the same concepts. Click the IDs or use them to filter the workbook.

P01 From a balance sheet to valuation

Where does a receivable end up in company value?

T04.01 T06.04 A02 T04.05 A05 A06 A11 A13 A19 A15 A16

Predict - inspect - explain - connect

P03 Follow a machine through the statements

Why is buying a machine different from recording depreciation?

T04.08 A07 T06.07 A08 A09 A10 A11 T16.06

Predict - inspect - explain - connect

P05 Is growth creating value?

Can a faster-growing company be worth less?

T11.15 T11.12 T17.11 A20 A13 T17.12 T17.18

Predict - inspect - explain - connect

P02 The profitable company with no cash

How can growing sales create a funding gap?

T05.01 T04.05 A02 T14.03 T14.02 T07.03 T14.04 T14.15

Predict - inspect - explain - connect

P04 Who gets the cash: firm versus equity

Why is net borrowing in FCFE but outside operating FCFF?

T17.03 A11 A12 A18 A13 A14 T17.04

Predict - inspect - explain - connect

P06 Read and challenge reported earnings

Which reported number needs further investigation?

T04.16 T04.06 T10.02 T10.06 T10.07 T10.11 T10.15

Predict - inspect - explain - connect

Question-led journeys through the web

Each route reuses the same concepts. Click the IDs or use them to filter the workbook.

P07 From business drivers to a model

If collections slow, what changes in the model?

T11.15 T15.06 T16.04 T16.05 T16.06 T16.07 T16.03 T16.16

Predict - inspect - explain - connect

P09 Build a portfolio for a real goal

Does the best standalone investment make the best portfolio?

T30.06 T22.01 T22.02 T22.04 T22.09 T22.12 T22.13 T22.15

Predict - inspect - explain - connect

P11 Startup runway and founder ownership

How much cash and ownership remain after a funding round?

T27.02 T27.03 T27.05 T27.07 T27.06 T27.09 T27.10 T27.17

Predict - inspect - explain - connect

P08 A bond meets a rate shock

Why does bond price change when yields change?

T03.01 T20.03 T20.05 T20.06 T20.10 T20.11 T20.16

Predict - inspect - explain - connect

P10 A bank lends and absorbs losses

How do lending losses connect to funding and capital?

T24.03 T24.04 T24.09 T24.12 A17 T24.07 T24.08 T25.07

Predict - inspect - explain - connect

P12 Should we acquire this business?

Does EPS accretion mean the deal creates value?

T26.01 T26.05 T26.06 T26.07 T26.08 T26.09 T26.10 T26.19

Predict - inspect - explain - connect

Question-led journeys through the web

Each route reuses the same concepts. Click the IDs or use them to filter the workbook.

P13 Finance an infrastructure project

Which cash flows support debt repayment?

T28.10 T28.11 T28.13 T28.12 T28.15 T28.14 T28.16 T28.17

Predict - inspect - explain - connect

P15 Retirement in purchasing-power terms

Will the future income buy what you expect?

T30.06 T03.11 T32.02 T03.09 T30.10 T30.11 T29.15

Predict - inspect - explain - connect

P17 Climate exposure to valuation

How could a carbon price affect an exposed company?

T37.03 T37.05 T16.04 T16.10 T17.07 T37.02 T37.10

Predict - inspect - explain - connect

P14 Hedge a foreign-currency payment

What does a currency hedge remove and what remains?

T33.01 T33.02 T33.07 T21.01 T33.08 T21.19 T21.18

Predict - inspect - explain - connect

P16 Payments to fintech unit economics

Who earns fees and who takes risk in a digital payment?

T36.01 T36.02 T36.03 T36.05 T36.11 T36.17 T11.15

Predict - inspect - explain - connect

P18 A backtest that deserves belief

Was this information actually available when the trade was made?

T38.01 T38.02 T38.04 T38.09 T38.10 T38.11 T38.18

Predict - inspect - explain - connect

Question-led journeys through the web

Each route reuses the same concepts. Click the IDs or use them to filter the workbook.

P19 Personal protection and long-term wealth

How do protection, debt and investment choices fit together?

T30.01 T30.02 T30.03 T30.04 T30.07 T30.08 T30.15

Predict - inspect - explain - connect

P21 Compare companies without mixing claims

Why is EV/EBITDA different from P/E?

T17.03 T17.04 T18.01 T18.07 T18.03 T18.04 T18.05 T18.08

Predict - inspect - explain - connect

P23 Catch a decision bias

Would you change your view if the same evidence were reframed?

T35.03 T35.05 T35.06 T19.12 T35.13 T40.18

Predict - inspect - explain - connect

P20 Insurance promises and investment assets

Can the insurer meet claims when they fall due?

T29.01 T29.05 T29.07 T29.09 T29.12 T29.13 T29.14

Predict - inspect - explain - connect

P22 Policy decisions to company value

How does a policy-rate change reach a valuation?

T32.05 T32.06 T20.08 T13.03 T13.08 A13 T17.18

Predict - inspect - explain - connect

P24 Specialist valuation and public markets

Which valuation convention changes with the business model?

T09.15 T39.01 T39.03 T39.06 T18.15 T18.13 T18.09 T19.14

Predict - inspect - explain - connect

Foundations of finance

15 original topic units. Identify who supplies capital, who uses it, and how value and risk move.

TOPICS AND STABLE IDS

T01.01	What finance is: allocating capital across time and uncertainty
T01.02	Personal finance, corporate finance, public finance, and investment finance
T01.03	The financial system: households, businesses, governments, and intermediaries
T01.04	Real assets versus financial assets
T01.05	Income, expenses, assets, liabilities, equity, and wealth
T01.06	Stocks versus flows: a balance at a point in time versus activity during a period
T01.07	Saving, investing, borrowing, lending, speculation, and hedging
T01.08	Sources and uses of funds
T01.09	Liquidity, profitability, solvency, and financial flexibility
T01.10	Risk, return, opportunity cost, and required return
T01.11	Economic profit versus accounting profit
T01.12	Value creation versus value transfer
T01.13	Financial decisions under certainty, risk, and uncertainty
T01.14	Agency problems, incentives, information asymmetry, and moral hazard
T01.15	Arbitrage, the law of one price, and market efficiency

A QUESTION TO OPEN THE AREA

Who provides the money, who bears the risk, and who receives the benefit?

FOLLOW THESE CONNECTIONS

T21.05 > T01.15

Put-call parity is an arbitrage relationship under its assumptions.

T29.05 > T01.14

Underwriting addresses adverse selection and information asymmetry.

T34.11 > T01.14

Governance mechanisms address agency problems and accountability.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Mathematics and quantitative foundations

15 original topic units. Choose the right mathematical or statistical tool and interpret its result.

TOPICS AND STABLE IDS

T02.01	Percentages, percentage changes, percentage points, and basis points
T02.02	Ratios, proportions, weighted averages, and index numbers
T02.03	Simple interest and compound interest
T02.04	Arithmetic returns, geometric returns, and logarithmic returns
T02.05	CAGR and annualisation
T02.06	Exponents, logarithms, and growth equations
T02.07	Probability, conditional probability, and Bayes' theorem
T02.08	Expected value, variance, standard deviation, and covariance
T02.09	Correlation and its limitations
T02.10	Probability distributions, skewness, kurtosis, and tail risk
T02.11	Sampling, estimation, confidence intervals, and hypothesis testing
T02.12	Regression, statistical significance, and economic significance
T02.13	Time-series foundations, stationarity, autocorrelation, and volatility
T02.14	Optimisation, linear algebra, and calculus for finance
T02.15	Simulation, sensitivity analysis, and scenario analysis

A QUESTION TO OPEN THE AREA

Could a mathematically correct answer still be financially misleading?

FOLLOW THESE CONNECTIONS

[T15.12 > T02.14](#)

Capacity and product-mix decisions can use constrained optimisation.

[T21.06 > T02.07](#)

Binomial pricing uses possible future states and risk-neutral valuation.

[T02.09 > T22.04](#)

Correlation helps explain diversification benefits.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Time value of money

15 original topic units. Compare amounts received or paid at different times on a consistent basis.

TOPICS AND STABLE IDS

T03.01	Present value and future value
T03.02	Discounting and compounding
T03.03	Discount factors and cash-flow timelines
T03.04	Ordinary annuities and annuities due
T03.05	Growing annuities
T03.06	Perpetuities and growing perpetuities
T03.07	Uneven cash flows
T03.08	Nominal rates, effective rates, and continuous compounding
T03.09	Real versus nominal interest rates
T03.10	Loan amortisation and EMI calculations
T03.11	Sinking funds and savings accumulation
T03.12	Net present value
T03.13	Internal rate of return and modified IRR
T03.14	Money-weighted versus time-weighted returns
T03.15	Matching cash-flow frequency, currency, and discount rates

A QUESTION TO OPEN THE AREA

Would you prefer the money today or later, and what rate changes your answer?

FOLLOW THESE CONNECTIONS

T03.06 > T17.12

A growing perpetuity provides a common terminal-value formula.

T12.05 > T03.12

Incremental project cash flows are discounted to calculate project NPV.

T03.01 > T20.05

Present-value calculations provide the basic bond-pricing framework.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Accounting foundations

16 original topic units. Recognise and measure transactions before interpreting financial statements.

TOPICS AND STABLE IDS

T04.01	The accounting equation
T04.02	Double-entry bookkeeping
T04.03	Debits and credits
T04.04	Journals, ledgers, trial balances, and closing entries
T04.05	Cash accounting versus accrual accounting
T04.06	Revenue recognition and expense recognition
T04.07	Accruals, deferrals, prepayments, and provisions
T04.08	Capital expenditure versus operating expenditure
T04.09	Capitalisation versus immediate expensing
T04.10	Historical cost, fair value, and amortised cost
T04.11	Materiality, prudence, consistency, and going concern
T04.12	Accounting estimates, policy changes, and errors
T04.13	Reporting periods and comparative statements
T04.14	Standalone versus consolidated accounts
T04.15	Financial reporting frameworks: Ind AS, IFRS, and US GAAP
T04.16	Annual reports, interim reports, and regulatory filings

A QUESTION TO OPEN THE AREA

Which accounts change when this transaction happens?

FOLLOW THESE CONNECTIONS

T04.01 > T06.18

The accounting equation makes funding claims reconcile with the asset side.

T04.02 > T07.14

Double entry lets one transaction be traced through accounts and statements.

T04.09 > T05.10

Capitalising expenditure can replace an immediate expense with later depreciation or amortisation.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Income statement

20 original topic units. Trace revenue and expenses into the profit measures used in analysis.

TOPICS AND STABLE IDS

T05.01	Revenue, sales, and other operating income
T05.02	Gross versus net revenue recognition
T05.03	Sales returns, discounts, rebates, and indirect taxes
T05.04	Cost of goods sold and cost of services
T05.05	Inventory changes and production costs
T05.06	Gross profit and gross margin
T05.07	Employee expenses
T05.08	Selling, general, and administrative expenses
T05.09	Research and development expenses
T05.10	Depreciation, amortisation, and impairment
T05.11	EBITDA, EBIT, and operating profit
T05.12	Finance costs and interest income
T05.13	Other income and investment gains
T05.14	Exceptional items and non-recurring adjustments
T05.15	Profit before tax and income tax expense
T05.16	Continuing versus discontinued operations
T05.17	Net profit and profit attributable to shareholders
T05.18	Basic and diluted earnings per share
T05.19	Other comprehensive income and total comprehensive income
T05.20	Segment revenue and segment profitability

A QUESTION TO OPEN THE AREA

Can this change reported profit without changing cash today?

FOLLOW THESE CONNECTIONS

A17 > T05.17

Credit-loss expense can reduce reported profit without a contemporaneous cash payment.

A08 > A09

Depreciation reduces EBIT for depreciating operating assets.

A08 > A11

Non-cash depreciation deducted in operating profit is added back in the standard FCFF build.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Balance sheet

20 original topic units. Explain how assets are funded and how balances change through transactions.

TOPICS AND STABLE IDS

T06.01	Current versus non-current classification
T06.02	Cash and cash equivalents
T06.03	Restricted cash and liquid investments
T06.04	Trade receivables and expected credit losses
T06.05	Inventory: raw materials, work in progress, and finished goods
T06.06	Prepayments, advances, and other current assets
T06.07	Property, plant, and equipment
T06.08	Capital work in progress
T06.09	Right-of-use assets
T06.10	Intangible assets and goodwill
T06.11	Financial investments and investment property
T06.12	Deferred tax assets and liabilities
T06.13	Trade payables and accrued expenses
T06.14	Contract assets, contract liabilities, and deferred revenue
T06.15	Short-term and long-term borrowings
T06.16	Lease liabilities
T06.17	Provisions, pension obligations, and employee benefits
T06.18	Share capital, securities premium, and retained earnings
T06.19	Reserves, treasury shares, and non-controlling interests
T06.20	Contingent liabilities, guarantees, and off-balance-sheet commitments

A QUESTION TO OPEN THE AREA

Where did this balance come from, and what transaction would change it?

FOLLOW THESE CONNECTIONS

T04.01 > T06.18

The accounting equation makes funding claims reconcile with the asset side.

A02 > A05

Operating receivables are one component of operating working capital.

A02 > A17

Uncollected customer balances may require an expected credit-loss allowance.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Cash-flow statement and three-statement connections

16 original topic units. Reconcile accounting performance with actual cash movement.

TOPICS AND STABLE IDS

T07.01	Cash flow from operations, investing, and financing
T07.02	Direct versus indirect operating cash flow
T07.03	Reconciling profit to operating cash flow
T07.04	Non-cash expenses and non-cash income
T07.05	Working-capital adjustments
T07.06	Capital expenditure and asset disposal proceeds
T07.07	Acquisitions and investment transactions
T07.08	Debt issuance and repayment
T07.09	Equity issuance, buybacks, and dividends
T07.10	Interest and dividend classifications across reporting frameworks
T07.11	Foreign-exchange effects on cash
T07.12	Restricted cash and cash reconciliation
T07.13	Free cash flow and its different definitions
T07.14	How transactions move through all three statements
T07.15	Why profitable companies can run out of cash
T07.16	Why cash-rich companies can report losses

A QUESTION TO OPEN THE AREA

Why might cash move differently from reported profit?

FOLLOW THESE CONNECTIONS

T04.02 > T07.14

Double entry lets one transaction be traced through accounts and statements.

A05 > A06

Increases in operating working capital generally reduce operating cash flow in the indirect reconciliation.

A07 > A11

Cash operating capital expenditure is deducted when calculating standard FCFF.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Advanced accounting and reporting

16 original topic units. Understand recognition and measurement choices that change reported numbers.

TOPICS AND STABLE IDS

T08.01	Business combinations and purchase-price allocation
T08.02	Goodwill recognition and impairment testing
T08.03	Subsidiaries, associates, and joint ventures
T08.04	Consolidation and intercompany eliminations
T08.05	Equity-method accounting
T08.06	Lease accounting
T08.07	Financial instrument classification and measurement
T08.08	Hedge accounting
T08.09	Foreign-currency transactions and translation
T08.10	Share-based compensation
T08.11	Pension and post-employment accounting
T08.12	Deferred tax and tax-loss carryforwards
T08.13	Construction and long-term contract accounting
T08.14	Securitisation and derecognition
T08.15	Related-party transactions
T08.16	Accounting for restructuring, disposals, and discontinued operations

A QUESTION TO OPEN THE AREA

Would another accounting treatment change the result without changing the business?

FOLLOW THESE CONNECTIONS

T06.09 > T08.06

Right-of-use assets arise from applicable lease accounting.

T08.01 > T26.11

Acquisition accounting feeds the pro-forma merger model.

T08.04 > T04.14

Consolidation combines controlled entities and eliminates intercompany amounts.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Financial statement analysis

18 original topic units. Interpret performance using consistent denominators, periods and peer context.

TOPICS AND STABLE IDS

T09.01	Common-size statements
T09.02	Horizontal and vertical analysis
T09.03	Growth analysis and trend analysis
T09.04	Liquidity ratios
T09.05	Profitability and margin ratios
T09.06	Efficiency and asset-turnover ratios
T09.07	Leverage, coverage, and solvency ratios
T09.08	ROA, ROE, ROCE, and ROIC
T09.09	DuPont analysis
T09.10	Cash conversion and earnings-to-cash conversion
T09.11	Operating leverage and financial leverage
T09.12	Sustainable growth and internal growth
T09.13	Incremental returns on capital
T09.14	Segment analysis
T09.15	Peer benchmarking and industry-specific ratios
T09.16	Normalising financial statements
T09.17	Seasonality, cyclicity, and base effects
T09.18	Reconciling reported, adjusted, and underlying performance

A QUESTION TO OPEN THE AREA

Could this ratio improve while the underlying business becomes weaker?

FOLLOW THESE CONNECTIONS

T17.11 > A20

Reinvestment and incremental operating returns connect growth to value creation.

A20 > A13

ROIC above a comparable cost of capital is a value-creation diagnostic.

T12.15 > A20

Capital allocation can be assessed through operating returns and value created.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Earnings quality, forensic accounting, and fraud

15 original topic units. Test whether reported performance is supported by cash, disclosures and controls.

TOPICS AND STABLE IDS

T10.01	Earnings quality versus earnings growth
T10.02	Aggressive revenue recognition
T10.03	Channel stuffing and round-tripping
T10.04	Expense deferral and improper capitalisation
T10.05	Provision manipulation and reserve releases
T10.06	Accrual quality
T10.07	Receivables and inventory warning signs
T10.08	Cash-flow classification manipulation
T10.09	Related-party dealings and fund diversion
T10.10	Hidden leverage and guarantees
T10.11	Auditor opinions, qualifications, and emphasis-of-matter paragraphs
T10.12	Restatements and internal-control weaknesses
T10.13	Financial distress and fraud-screening models
T10.14	Management adjustments and misleading non-GAAP metrics
T10.15	Financial footnote investigation

A QUESTION TO OPEN THE AREA

What evidence would make you doubt the reported number?

FOLLOW THESE CONNECTIONS

[T10.02 > T04.06](#)

Revenue-quality analysis tests whether recognition reflects applicable criteria.

[T10.05 > T06.17](#)

Provision and reserve movements can affect reported earnings.

[T10.06 > T07.03](#)

Accrual quality is assessed partly by reconciling earnings to operating cash.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Business economics and unit economics

15 original topic units. Connect operating choices and customer economics to financial performance.

TOPICS AND STABLE IDS

T11.01	Revenue models and cost structures
T11.02	Fixed, variable, and semi-variable costs
T11.03	Contribution margin and break-even analysis
T11.04	Price-volume-mix analysis
T11.05	Capacity utilisation and operating leverage
T11.06	Economies of scale and economies of scope
T11.07	Customer acquisition cost
T11.08	Customer lifetime value
T11.09	Retention, churn, and cohort economics
T11.10	Customer and product profitability
T11.11	Payback periods and contribution after acquisition costs
T11.12	Working-capital intensity and capital intensity
T11.13	Network effects and financial implications
T11.14	Pricing, discounting, and revenue management
T11.15	Linking business drivers to financial results

A QUESTION TO OPEN THE AREA

Can selling more make the business less profitable?

FOLLOW THESE CONNECTIONS

T11.02 > T11.03

Fixed and variable costs determine contribution and break-even volume.

T11.04 > T05.01

Price, volume and mix explain changes in revenue.

T11.05 > T09.11

Fixed operating costs and utilisation contribute to operating leverage.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Corporate finance and capital allocation

15 original topic units. Compare uses of capital using incremental cash flows and risk.

TOPICS AND STABLE IDS

T12.01	Objectives of the firm and shareholder value
T12.02	Stakeholder trade-offs and agency costs
T12.03	Investment, financing, and payout decisions
T12.04	Capital budgeting
T12.05	Incremental cash flows
T12.06	Sunk costs, opportunity costs, and externalities
T12.07	Replacement, expansion, and mutually exclusive projects
T12.08	NPV, IRR, MIRR, payback, and profitability index
T12.09	Equivalent annual cost
T12.10	Capital rationing
T12.11	Project risk and scenario analysis
T12.12	Real options: delay, expand, abandon, and switch
T12.13	Dividends and share repurchases
T12.14	Organic growth versus acquisitions
T12.15	Capital allocation performance and management incentives

A QUESTION TO OPEN THE AREA

Would this decision create value after allowing for risk and opportunity cost?

FOLLOW THESE CONNECTIONS

T12.05 > T03.12

Incremental project cash flows are discounted to calculate project NPV.

T12.06 > T12.05

Relevant cash flows include opportunity costs but exclude sunk costs.

T12.12 > T21.02

Managerial flexibility has option-like payoffs.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Cost of capital and capital structure

18 original topic units. Match financing risk, capital claims and required returns.

TOPICS AND STABLE IDS

T13.01	Cost of equity
T13.02	Capital Asset Pricing Model
T13.03	Risk-free rate and equity risk premium
T13.04	Beta estimation
T13.05	Levered and unlevered beta
T13.06	Bottom-up beta
T13.07	Country risk and currency consistency
T13.08	Cost of debt and credit spreads
T13.09	Preferred capital and hybrid instruments
T13.10	Weighted average cost of capital
T13.11	Market-value versus book-value weights
T13.12	Marginal versus historical financing costs
T13.13	Modigliani-Miller propositions
T13.14	Interest tax shields
T13.15	Trade-off theory and pecking-order theory
T13.16	Financial distress costs and optimal leverage
T13.17	Adjusted present value
T13.18	Changing leverage and project-specific discount rates

A QUESTION TO OPEN THE AREA

Which investor claim bears this risk and which return compensates for it?

FOLLOW THESE CONNECTIONS

A11 > A13

FCFF is discounted at a matching WACC in a standard enterprise DCF.

A20 > A13

ROIC above a comparable cost of capital is a value-creation diagnostic.

T31.03 > T13.14

The tax benefit of interest depends on tax capacity and deduction rules.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Working capital and treasury management

16 original topic units. Manage the timing gap between cash receipts, payments and funding.

TOPICS AND STABLE IDS

T14.01	Operating working capital versus total working capital
T14.02	Cash conversion cycle
T14.03	Inventory days, receivable days, and payable days
T14.04	Cash budgeting and short-term liquidity forecasting
T14.05	Customer credit policy and collections
T14.06	Inventory financing
T14.07	Supplier terms and payment management
T14.08	Factoring and invoice discounting
T14.09	Reverse factoring and supply-chain finance
T14.10	Cash-credit facilities, overdrafts, and revolving credit
T14.11	Commercial paper and short-term funding
T14.12	Cash pooling and intercompany funding
T14.13	Treasury investment policies
T14.14	Foreign-exchange and interest-rate exposure
T14.15	Liquidity buffers and contingency funding
T14.16	Treasury controls and counterparty management

A QUESTION TO OPEN THE AREA

Can a growing business run out of cash before customers pay?

FOLLOW THESE CONNECTIONS

A02 > A05

Operating receivables are one component of operating working capital.

A05 > A06

Increases in operating working capital generally reduce operating cash flow in the indirect reconciliation.

T14.03 > T14.02

Inventory days plus receivable days less payable days gives the cash conversion cycle.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Financial planning, budgeting, and management accounting

15 original topic units. Translate operating assumptions into plans, decisions and explanations of variance.

TOPICS AND STABLE IDS

T15.01	Strategic financial planning
T15.02	Annual operating plans
T15.03	Master budgets and departmental budgets
T15.04	Zero-based, incremental, and activity-based budgeting
T15.05	Flexible budgets and rolling forecasts
T15.06	Revenue, cost, headcount, and capital expenditure forecasting
T15.07	Variance analysis
T15.08	Standard costing and activity-based costing
T15.09	Cost allocation and responsibility centres
T15.10	Relevant costing and make-or-buy decisions
T15.11	Transfer pricing for internal decisions
T15.12	Product-mix and capacity decisions
T15.13	Management dashboards and KPI trees
T15.14	Forecast accuracy
T15.15	Business partnering and board reporting

A QUESTION TO OPEN THE AREA

Which operating assumption explains the gap between plan and actual results?

FOLLOW THESE CONNECTIONS

T15.06 > T16.04
Operating forecasts supply model assumptions and schedules.

T15.07 > T15.14
Variance analysis diagnoses differences between forecasts and actual results.

T15.08 > T11.10
Costing methods affect measured product and customer profitability.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Financial modelling

16 original topic units. Connect schedules into an internally consistent and auditable model.

TOPICS AND STABLE IDS

T16.01	Model structure, assumptions, and conventions
T16.02	Historical financial statement preparation
T16.03	Three-statement models
T16.04	Revenue and operating-cost schedules
T16.05	Working-capital schedules
T16.06	Fixed-asset and depreciation schedules
T16.07	Debt, interest, and cash schedules
T16.08	Tax and equity schedules
T16.09	Circularity and model balancing
T16.10	Scenario and sensitivity models
T16.11	DCF and comparable-company models
T16.12	Merger and acquisition models
T16.13	Leveraged buyout models
T16.14	Project finance and real estate models
T16.15	Bank, insurance, and startup models
T16.16	Model auditing, error checks, and documentation

A QUESTION TO OPEN THE AREA

If this input changes, which schedules should change with it?

FOLLOW THESE CONNECTIONS

T11.15 > T16.04

Operating drivers become revenue and cost schedules in a financial model.

T15.06 > T16.04

Operating forecasts supply model assumptions and schedules.

T16.03 > T07.14

A three-statement model connects transaction effects across the statements.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Valuation foundations and discounted cash flow

19 original topic units. Match cash flows, reinvestment, discount rates and investor claims.

TOPICS AND STABLE IDS

T17.01	Price versus value
T17.02	Intrinsic valuation versus relative valuation
T17.03	Enterprise value versus equity value
T17.04	The enterprise-to-equity value bridge
T17.05	FCFF and FCFE
T17.06	Dividend discount models
T17.07	Discounted cash-flow valuation
T17.08	Adjusted present value
T17.09	Residual income and economic profit valuation
T17.10	Forecast horizons and competitive advantage periods
T17.11	Reinvestment, growth, and return on capital
T17.12	Terminal growth and terminal value
T17.13	Exit-multiple terminal values
T17.14	Mid-year discounting and stub periods
T17.15	Excess cash and non-operating assets
T17.16	Debt, leases, pension deficits, and minority interests
T17.17	Share dilution, options, and restricted stock
T17.18	Sensitivity analysis, scenarios, and reverse DCF
T17.19	Valuation ranges and uncertainty

A QUESTION TO OPEN THE AREA

Does this valuation use the right cash flow for the right investor?

FOLLOW THESE CONNECTIONS

T17.16 > T17.04

Non-equity claims provide deductions in a consistent enterprise-to-equity bridge.

A08 > A11

Non-cash depreciation deducted in operating profit is added back in the standard FCFF build.

A07 > A11

Cash operating capital expenditure is deducted when calculating standard FCFF.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Relative valuation and specialist valuation

18 original topic units. Compare value using economically comparable businesses and consistent measures.

TOPICS AND STABLE IDS

T18.01	Comparable-company selection
T18.02	Comparable transaction analysis
T18.03	Trailing versus forward multiples
T18.04	P/E, PEG, P/B, and P/S
T18.05	EV/revenue, EV/EBITDA, and EV/EBIT
T18.06	Cash-flow and sector-specific multiples
T18.07	Normalising earnings and accounting differences
T18.08	Growth, risk, margins, and returns as multiple drivers
T18.09	Sum-of-the-parts valuation
T18.10	Net asset value and liquidation value
T18.11	Replacement cost
T18.12	Private-company valuation
T18.13	Startup and pre-revenue valuation
T18.14	Cyclical, commodity, and distressed-company valuation
T18.15	Bank, NBFC, and insurance valuation
T18.16	Intangible assets, brands, patents, and customer relationships
T18.17	Control premiums and marketability considerations
T18.18	Business valuation versus transaction pricing

A QUESTION TO OPEN THE AREA

Are these businesses comparable enough for this multiple to mean anything?

FOLLOW THESE CONNECTIONS

[T18.01 > T18.07](#)

Peer selection requires accounting and economic comparability.

[T05.18 > T18.04](#)

Earnings per share is the denominator of a share-price P/E ratio.

[T06.18 > T18.04](#)

Equity book value contributes to price-to-book comparisons.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Equity markets and equity research

17 original topic units. Translate business analysis into an investment thesis and understand trading mechanics.

TOPICS AND STABLE IDS

T19.01	Common shares, preference shares, and depositary receipts
T19.02	Primary and secondary equity markets
T19.03	IPOs, follow-on offerings, rights issues, and private placements
T19.04	Stock exchanges, indices, and market capitalisation
T19.05	Free float and index construction
T19.06	Dividends, splits, bonus issues, and buybacks
T19.07	Order types and execution basics
T19.08	Margin trading and short selling
T19.09	Securities lending
T19.10	Fundamental analysis
T19.11	Industry analysis and competitive advantage
T19.12	Investment theses, catalysts, and risk factors
T19.13	Earnings estimates and earnings surprises
T19.14	Equity research reports
T19.15	Growth, value, quality, momentum, and income investing
T19.16	Technical analysis and its limitations
T19.17	Shareholding patterns and promoter pledging

A QUESTION TO OPEN THE AREA

What would have to happen for your investment thesis to be wrong?

FOLLOW THESE CONNECTIONS

T17.01 > T19.12

Comparing market price and estimated value shapes an investment thesis.

T18.03 > T19.13

Forward multiples use forward earnings or other forecast measures.

T12.13 > T19.06

Payout decisions are implemented through dividends and share repurchases.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Fixed income and debt markets

17 original topic units. Relate debt cash flows, yields, interest-rate sensitivity and credit quality.

TOPICS AND STABLE IDS

T20.01	Government securities, treasury bills, and corporate bonds
T20.02	Municipal, agency, covered, and inflation-linked bonds
T20.03	Coupon structures and payment conventions
T20.04	Clean price, dirty price, and accrued interest
T20.05	Bond pricing
T20.06	Current yield, yield to maturity, yield to call, and yield to worst
T20.07	Spot rates, forward rates, and par yields
T20.08	Yield curves and term-structure theories
T20.09	Bootstrapping discount curves
T20.10	Duration, modified duration, and effective duration
T20.11	Convexity and key-rate duration
T20.12	Credit spreads and spread duration
T20.13	Floating-rate notes
T20.14	Callable, puttable, and convertible bonds
T20.15	Default, recovery, and bond covenants
T20.16	Bond portfolio strategies, ladders, and immunisation
T20.17	Fixed-income attribution and relative value

A QUESTION TO OPEN THE AREA

Which matters more here: a change in yields, credit quality or payment timing?

FOLLOW THESE CONNECTIONS

T20.03 > T20.05

Coupon and principal conventions define bond cash flows.

T03.01 > T20.05

Present-value calculations provide the basic bond-pricing framework.

T20.12 > T24.11

Credit spreads reflect default risk along with liquidity and other premiums.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Derivatives and structured products

19 original topic units. Understand contingent payoffs, pricing inputs and hedge limitations.

TOPICS AND STABLE IDS

T21.01	Forwards and futures
T21.02	Options: calls and puts
T21.03	Payoff versus profit diagrams
T21.04	Moneyness, intrinsic value, and time value
T21.05	Put-call parity
T21.06	Binomial option pricing
T21.07	Black-Scholes foundations and assumptions
T21.08	Option Greeks
T21.09	Implied versus realised volatility
T21.10	Volatility smiles, skews, and term structures
T21.11	Option strategies: spreads, straddles, strangles, and collars
T21.12	Interest-rate, currency, and commodity swaps
T21.13	Credit default swaps
T21.14	Caps, floors, and swaptions
T21.15	Exotic options and path-dependent payoffs
T21.16	Structured notes and embedded derivatives
T21.17	Hedging, speculation, and arbitrage
T21.18	Margin, collateral, clearing, and counterparty risk
T21.19	Basis risk and hedge effectiveness

A QUESTION TO OPEN THE AREA

What is the payoff if the market moves against your position?

FOLLOW THESE CONNECTIONS

[T12.12 > T21.02](#)

Managerial flexibility has option-like payoffs.

[T20.14 > T21.02](#)

Callable, puttable and convertible bonds contain option-like features.

[T21.01 > T33.08](#)

Currency forwards and futures can hedge specified currency exposures.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Portfolio management and asset allocation

19 original topic units. Combine assets to meet goals and evaluate risk-adjusted performance.

TOPICS AND STABLE IDS

T22.01	Investment objectives and constraints
T22.02	Investment policy statements
T22.03	Risk tolerance, risk capacity, and investment horizon
T22.04	Diversification and correlation
T22.05	Modern portfolio theory
T22.06	Efficient frontier and minimum-variance portfolios
T22.07	Capital allocation line and capital market line
T22.08	CAPM, arbitrage pricing theory, and factor models
T22.09	Strategic and tactical asset allocation
T22.10	Active versus passive management
T22.11	Indexing, tracking error, and active share
T22.12	Portfolio construction and position sizing
T22.13	Rebalancing
T22.14	Benchmark selection
T22.15	Sharpe, Sortino, Treynor, and information ratios
T22.16	Alpha and performance attribution
T22.17	Drawdowns and downside protection
T22.18	Liability-driven and goals-based investing
T22.19	Endowment, pension, and institutional portfolios

A QUESTION TO OPEN THE AREA

Would this investment improve the portfolio rather than just look attractive alone?

FOLLOW THESE CONNECTIONS

T22.09 > T22.12

Asset allocation provides target exposures and constraints for portfolio construction.

T19.05 > T22.11

Index membership and weights affect index-tracking portfolios.

T20.16 > T22.18

Immunisation and bond ladders can support liability-oriented portfolios.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Investment vehicles and alternative assets

19 original topic units. Understand asset exposure, fund structures, fees, liquidity and measured performance.

TOPICS AND STABLE IDS

T23.01	Mutual funds and exchange-traded funds
T23.02	Open-end and closed-end funds
T23.03	Index funds and actively managed funds
T23.04	Fund NAV, expense ratios, and tracking differences
T23.05	Fund selection and manager due diligence
T23.06	Portfolio management services and alternative investment funds
T23.07	Hedge funds and hedge-fund strategies
T23.08	Private equity
T23.09	Venture capital and growth equity
T23.10	Private credit and direct lending
T23.11	Distressed debt and special situations
T23.12	Real estate funds, REITs, and InvITs
T23.13	Commodities, gold, and natural resources
T23.14	Infrastructure funds
T23.15	Collectibles and other alternative stores of value
T23.16	Fund fees, carried interest, and distribution waterfalls
T23.17	Capital calls, commitments, and the J-curve
T23.18	IRR, MOIC, DPI, RVPI, and TVPI
T23.19	Liquidity, valuation, and survivorship bias in private assets

A QUESTION TO OPEN THE AREA

How do fees, illiquidity and the fund structure change the investor outcome?

FOLLOW THESE CONNECTIONS

T23.04 > T23.01

NAV, expense ratios and tracking differences help assess fund implementation and investor outcomes.

T22.10 > T23.03

Active and passive choices can be implemented through suitable funds.

T23.04 > T30.18

Fees affect investor outcomes and adviser evaluation.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Banking and credit analysis

19 original topic units. Connect credit origination, bank funding, loan losses and regulatory capital.

TOPICS AND STABLE IDS

T24.01	How banks create and intermediate credit
T24.02	Bank balance sheets and income statements
T24.03	Deposits, lending, and wholesale funding
T24.04	Net interest income and net interest margin
T24.05	Fee income and operating efficiency
T24.06	Asset-liability management
T24.07	Capital adequacy and risk-weighted assets
T24.08	Liquidity and funding risk
T24.09	Loan origination and underwriting
T24.10	Retail, SME, and corporate lending
T24.11	Credit ratings and credit scoring
T24.12	Probability of default, loss given default, and exposure at default
T24.13	Collateral, guarantees, and security structures
T24.14	Covenants and debt service capacity
T24.15	Non-performing assets and provisioning
T24.16	Expected credit losses
T24.17	Loan restructuring and recovery
T24.18	NBFCs, microfinance, and housing finance
T24.19	Securitisation and loan sales

A QUESTION TO OPEN THE AREA

Could this borrower repay under a plausible downside scenario?

FOLLOW THESE CONNECTIONS

A02 > A17

Uncollected customer balances may require an expected credit-loss allowance.

A17 > T05.17

Credit-loss expense can reduce reported profit without a contemporaneous cash payment.

T13.16 > T24.11

Credit quality and leverage help assess distress risk and financing spreads.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Financial risk management

16 original topic units. Identify exposures, measure losses and design governance and controls.

TOPICS AND STABLE IDS

T25.01	Market, credit, liquidity, and operational risk
T25.02	Business, strategic, and reputational risk
T25.03	Counterparty and settlement risk
T25.04	Concentration and contagion risk
T25.05	Value at Risk and expected shortfall
T25.06	Historical, parametric, and Monte Carlo methods
T25.07	Stress testing and reverse stress testing
T25.08	Scenario design
T25.09	Risk limits, risk appetite, and risk governance
T25.10	Credit portfolio risk
T25.11	Asset-liability mismatches
T25.12	Model risk and validation
T25.13	Operational losses, cyber risk, and fraud risk
T25.14	Enterprise risk management
T25.15	Economic capital and risk-adjusted performance
T25.16	Systemic risk and financial crises

A QUESTION TO OPEN THE AREA

Which exposure is missing from the headline risk number?

FOLLOW THESE CONNECTIONS

T19.17 > T25.04

Ownership concentration and pledged shares can introduce specific governance or financing risks.

T22.17 > T25.07

Stress scenarios help explore drawdowns and protection limits.

T24.07 > T25.15

Regulatory capital and economic capital have related but distinct objectives.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Mergers, acquisitions, and restructuring

19 original topic units. Separate operating value, financing, deal terms and execution risk.

TOPICS AND STABLE IDS

T26.01	Strategic rationale for acquisitions
T26.02	Horizontal, vertical, and conglomerate transactions
T26.03	Buy-side and sell-side processes
T26.04	Target screening
T26.05	Financial, commercial, tax, and legal due diligence
T26.06	Transaction valuation
T26.07	Synergies, dis-synergies, and integration costs
T26.08	Cash, stock, and mixed consideration
T26.09	Acquisition financing
T26.10	Accretion and dilution
T26.11	Merger models
T26.12	Purchase-price adjustments and working-capital mechanisms
T26.13	Earn-outs and contingent consideration
T26.14	Leveraged and management buyouts
T26.15	Divestitures, carve-outs, spin-offs, and split-offs
T26.16	Takeover defences and shareholder activism
T26.17	Financial restructuring and debt exchanges
T26.18	Bankruptcy, insolvency, and recovery waterfalls
T26.19	Post-merger integration and value tracking

A QUESTION TO OPEN THE AREA

Is the proposed value coming from the business, synergies or financing?

FOLLOW THESE CONNECTIONS

T18.02 > T26.06

Comparable deals provide evidence for transaction valuation.

T08.01 > T26.11

Acquisition accounting feeds the pro-forma merger model.

T12.14 > T26.01

Acquisitions and organic investment are alternative routes for deploying capital.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Startup and entrepreneurial finance

17 original topic units. Connect operating milestones, cash runway, financing terms and founder ownership.

TOPICS AND STABLE IDS

T27.01	Bootstrapping and startup cash management
T27.02	Business models and startup financial plans
T27.03	Burn rate, net burn, and runway
T27.04	Product-market fit and financing milestones
T27.05	Angel, seed, venture, and growth funding
T27.06	Cap tables and dilution
T27.07	Pre-money and post-money valuation
T27.08	Convertible notes and SAFE-type instruments
T27.09	Term sheets
T27.10	Liquidation preferences and participation rights
T27.11	Anti-dilution provisions
T27.12	Founder vesting and employee stock options
T27.13	Venture debt
T27.14	Fundraising strategy and investor communication
T27.15	SaaS, marketplace, consumer, and fintech economics
T27.16	Down rounds, bridge rounds, and recapitalisation
T27.17	Exit routes and exit waterfalls

A QUESTION TO OPEN THE AREA

How long can this startup operate and who owns it after the next round?

FOLLOW THESE CONNECTIONS

T19.03 > T27.17
An IPO can provide a startup or private investor exit route.

T23.09 > T27.05
Venture and growth investors finance startup funding rounds.

T26.08 > T27.06
Stock consideration can change ownership and dilution.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Real estate, infrastructure, and project finance

18 original topic units. Link asset or project cash flows, debt service and investor distributions.

TOPICS AND STABLE IDS

T28.01	Property types and real estate investment models
T28.02	Rental income, vacancy, and operating expenses
T28.03	Net operating income and capitalisation rates
T28.04	Property valuation
T28.05	Development feasibility and construction cash flows
T28.06	Mortgage structures and amortisation
T28.07	Loan-to-value, debt yield, and debt service coverage
T28.08	Levered and unlevered property returns
T28.09	Real estate investment waterfalls
T28.10	Project finance versus corporate finance
T28.11	Special-purpose vehicles and limited-recourse structures
T28.12	Construction risk and operating risk
T28.13	Concessions, offtake agreements, and public-private partnerships
T28.14	Debt sizing and debt sculpting
T28.15	DSCR, LLCR, and PLCR
T28.16	Reserve accounts and cash waterfalls
T28.17	Refinancing and project equity returns
T28.18	Energy, transport, utilities, and social infrastructure finance

A QUESTION TO OPEN THE AREA

Does this asset generate enough cash when the debt payments fall due?

FOLLOW THESE CONNECTIONS

T16.14 > T28.14

Project models support debt sizing and repayment sculpting.

T23.12 > T28.04

Real estate funds and REITs depend on property economics and valuation.

T28.02 > T28.03

Rental cash economics contribute to property NOI.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Insurance and actuarial finance

16 original topic units. Connect uncertain claims, premiums, reserves, investments and solvency.

TOPICS AND STABLE IDS

T29.01	Risk pooling and the economics of insurance
T29.02	Life, health, general, and liability insurance
T29.03	Term versus savings-linked insurance products
T29.04	Premiums, deductibles, exclusions, and claims
T29.05	Underwriting and adverse selection
T29.06	Reinsurance
T29.07	Claims reserves and technical provisions
T29.08	Mortality, morbidity, frequency, and severity
T29.09	Present value of insurance liabilities
T29.10	Loss ratios, expense ratios, and combined ratios
T29.11	Embedded value and value of new business
T29.12	Insurance investment portfolios
T29.13	Asset-liability matching
T29.14	Solvency and regulatory capital
T29.15	Pension and annuity products
T29.16	Catastrophe risk and insurance-linked securities

A QUESTION TO OPEN THE AREA

What happens if claims arrive sooner or cost more than expected?

FOLLOW THESE CONNECTIONS

T29.01 > T02.07
Insurance pools uncertain claim events using probability and exposure assumptions.

T29.05 > T01.14
Underwriting addresses adverse selection and information asymmetry.

T29.06 > T29.16
Reinsurance can transfer parts of catastrophe and other insured risks.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Personal finance and wealth management

18 original topic units. Connect household cash flow, goals, protection, investing and long-term wealth.

TOPICS AND STABLE IDS

T30.01	Personal financial statements and net worth
T30.02	Budgeting and expense tracking
T30.03	Emergency funds
T30.04	Debt management and credit scores
T30.05	Education, vehicle, and home loans
T30.06	Financial goals and savings plans
T30.07	Insurance needs analysis
T30.08	Investment selection and asset allocation
T30.09	Tax-efficient investing
T30.10	Retirement planning and withdrawal strategies
T30.11	Inflation and longevity risk
T30.12	Employee compensation and stock benefits
T30.13	Property ownership versus renting
T30.14	Family financial planning
T30.15	Estate planning, nominations, wills, and trusts
T30.16	Wealth preservation and intergenerational transfer
T30.17	Financial scams and investor protection
T30.18	Adviser selection, fees, and conflicts of interest

A QUESTION TO OPEN THE AREA

Does this choice move you closer to your goal without weakening protection?

FOLLOW THESE CONNECTIONS

T30.06 > T30.08

Financial goals and their timing inform personal investment selection and asset allocation.

T22.03 > T30.10

Risk capacity and horizon affect retirement investment choices.

T23.04 > T30.18

Fees affect investor outcomes and adviser evaluation.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Taxation in finance

16 original topic units. Distinguish accounting outcomes from tax outcomes and compare after-tax cash flows.

TOPICS AND STABLE IDS

T31.01	Direct versus indirect taxation
T31.02	Taxable income versus accounting income
T31.03	Marginal versus effective tax rates
T31.04	Individual and corporate taxation
T31.05	Capital gains, dividends, and interest taxation
T31.06	Tax treatment of investment products
T31.07	Tax depreciation and capital allowances
T31.08	Loss set-offs and carryforwards
T31.09	Deferred tax
T31.10	GST and its working-capital implications
T31.11	Withholding taxes and tax deducted at source
T31.12	Cross-border taxation and tax treaties
T31.13	Transfer pricing for tax purposes
T31.14	Transaction and restructuring taxation
T31.15	Tax planning versus tax evasion
T31.16	After-tax returns and tax-aware valuation

A QUESTION TO OPEN THE AREA

What changes when you compare after-tax cash rather than accounting profit?

FOLLOW THESE CONNECTIONS

T31.02 > T31.09

Differences between tax and book recognition produce some deferred tax effects.

T31.07 > A10

Tax depreciation affects operating cash taxes separately from book depreciation.

T31.03 > T13.14

The tax benefit of interest depends on tax capacity and deduction rules.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Macroeconomics, monetary policy, and public finance

16 original topic units. Trace economic conditions and policy transmission into funding, earnings and asset prices.

TOPICS AND STABLE IDS

T32.01	GDP, growth, productivity, and business cycles
T32.02	Inflation, disinflation, and deflation
T32.03	Employment, wages, and consumption
T32.04	Money supply, credit, and liquidity
T32.05	Central banks and monetary transmission
T32.06	Policy rates, reserve requirements, and open-market operations
T32.07	Quantitative easing and tightening
T32.08	Fiscal policy, taxation, and government spending
T32.09	Fiscal deficits and public debt
T32.10	Sovereign borrowing and debt sustainability
T32.11	Balance of payments and international capital flows
T32.12	Exchange-rate regimes
T32.13	Sovereign credit risk
T32.14	Financial repression and capital controls
T32.15	Leading indicators and macroeconomic forecasting
T32.16	How macroeconomic changes affect companies, valuations, and portfolios

A QUESTION TO OPEN THE AREA

Through which channel would this policy change affect a company or portfolio?

FOLLOW THESE CONNECTIONS

T20.13 > T32.06

Floating coupons reset against specified reference-rate conventions.

T25.07 > T32.16

Macro shocks can be translated into financial stress scenarios.

T32.02 > T03.09

Inflation separates nominal and real returns.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

International finance and foreign exchange

16 original topic units. Keep currency exposure, cash flows, financing and discount rates consistent.

TOPICS AND STABLE IDS

T33.01	Currency quotations and cross-rates
T33.02	Spot and forward foreign-exchange markets
T33.03	Purchasing power parity
T33.04	Covered and uncovered interest-rate parity
T33.05	Forward premiums and discounts
T33.06	Carry trades
T33.07	Transaction, translation, and economic exposure
T33.08	Currency hedging
T33.09	Cross-border investment and financing
T33.10	International cost of capital
T33.11	Country and political risk
T33.12	Multinational capital budgeting
T33.13	Trade finance, letters of credit, and documentary collections
T33.14	Export credit and international guarantees
T33.15	Offshore funding and foreign-currency debt
T33.16	Global portfolio diversification

A QUESTION TO OPEN THE AREA

What happens if the currency moves against the cash flow you expect?

FOLLOW THESE CONNECTIONS

T13.07 > T33.10

Country risk and currency choice affect an international cost-of-capital estimate.

T21.01 > T33.08

Currency forwards and futures can hedge specified currency exposures.

T31.12 > T33.09

Cross-border tax rules affect international investment and financing cash flows.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Market infrastructure, regulation, and governance

15 original topic units. Understand how transactions are executed, safeguarded and governed.

TOPICS AND STABLE IDS

T34.01	Exchanges, brokers, dealers, and market makers
T34.02	Custodians, depositories, and clearing houses
T34.03	Trade execution, clearing, and settlement
T34.04	Order books, spreads, depth, and liquidity
T34.05	Market impact and transaction costs
T34.06	Auctions, circuit breakers, and trading halts
T34.07	Securities issuance and disclosure requirements
T34.08	Insider trading and market manipulation
T34.09	Suitability, fiduciary duties, and investor protection
T34.10	Know-your-customer and anti-money-laundering frameworks
T34.11	Corporate boards, audit committees, and governance
T34.12	Shareholder rights and stewardship
T34.13	Credit-rating agencies and conflicts of interest
T34.14	Financial ethics and professional conduct
T34.15	Regulatory reporting and compliance

A QUESTION TO OPEN THE AREA

What protects the investor between placing a trade and receiving the asset?

FOLLOW THESE CONNECTIONS

T19.04 > T34.01
Equity markets operate through exchanges and intermediaries.

T21.18 > T34.02
Clearing and collateral mitigate some counterparty exposure.

T25.09 > T34.11
Boards and governance structures oversee risk appetite and limits.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Behavioural finance and financial decision-making

14 original topic units. Recognise how incentives and cognitive biases influence financial decisions.

TOPICS AND STABLE IDS

T35.01	Expected utility and risk preferences
T35.02	Prospect theory and loss aversion
T35.03	Anchoring and framing
T35.04	Mental accounting
T35.05	Overconfidence and self-attribution
T35.06	Confirmation and availability biases
T35.07	Herding and social influence
T35.08	Recency bias and extrapolation
T35.09	Disposition effect
T35.10	Present bias and self-control
T35.11	Bubbles, panics, and speculative cycles
T35.12	Behavioural corporate finance
T35.13	Decision journals, checklists, and pre-mortems
T35.14	Incentive design and decision quality

A QUESTION TO OPEN THE AREA

Would you make the same choice if the information were framed differently?

FOLLOW THESE CONNECTIONS

T35.02 > T22.03

Loss aversion can influence stated risk preferences and investment behaviour.

T35.03 > T17.18

Anchoring can distort valuation assumptions and scenario interpretation.

T35.04 > T30.02

Mental accounting can influence household budget and spending choices.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Fintech, payments, and digital finance

17 original topic units. Connect financial infrastructure, digital distribution, unit economics and controls.

TOPICS AND STABLE IDS

T36.01	Payment systems and payment processing
T36.02	Cards, acquiring, issuing, and interchange
T36.03	UPI and account-to-account payments
T36.04	Wallets, remittances, and cross-border payments
T36.05	Payment gateways and aggregators
T36.06	Digital lending and alternative underwriting
T36.07	Embedded finance and banking-as-a-service
T36.08	Open banking and account aggregation
T36.09	Robo-advisory and automated wealth management
T36.10	Insurtech, regtech, and wealthtech
T36.11	Fraud detection and financial identity
T36.12	Blockchain and distributed ledgers
T36.13	Cryptoassets and token economics
T36.14	Stablecoins and central bank digital currencies
T36.15	Decentralised finance, lending, and automated market makers
T36.16	Tokenisation of financial assets
T36.17	Fintech business models, unit economics, and risks

A QUESTION TO OPEN THE AREA

Who earns money from the transaction and who carries the credit or fraud risk?

FOLLOW THESE CONNECTIONS

T25.13 > T36.11

Identity and fraud controls address selected digital operational risks.

T34.10 > T36.11

Financial identity checks and monitoring support anti-fraud and AML controls.

T36.02 > T11.01

Card payments distribute fees and responsibilities across participants.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Sustainable finance and specialised financial systems

14 original topic units. Connect financing decisions with sustainability, inclusion and alternative structures.

TOPICS AND STABLE IDS

T37.01	ESG integration in investment decisions
T37.02	Sustainability reporting and materiality
T37.03	Climate transition risk and physical risk
T37.04	Green bonds and sustainability-linked finance
T37.05	Carbon markets and carbon pricing
T37.06	Impact investing and impact measurement
T37.07	Blended finance and development finance
T37.08	Social finance and financial inclusion
T37.09	Stewardship and engagement
T37.10	Greenwashing and data-quality limitations
T37.11	Islamic finance principles
T37.12	Murabaha, ijara, musharaka, mudaraba, and sukuk
T37.13	Cooperative finance and community finance
T37.14	Nonprofit and endowment finance

A QUESTION TO OPEN THE AREA

How would you distinguish a measurable impact from a financing label?

FOLLOW THESE CONNECTIONS

[T34.12 > T37.09](#)

Stewardship includes exercising rights and engaging with investees.

[T37.01 > T22.02](#)

ESG objectives and risks can be reflected in investment policy and analysis.

[T37.02 > T04.11](#)

Sustainability reporting also requires materiality judgements.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Computational finance, financial data, and AI

18 original topic units. Produce reproducible financial analysis without data leakage or model misuse.

TOPICS AND STABLE IDS

T38.01	Financial data sources and data licensing
T38.02	Data cleaning and reconciliation
T38.03	Corporate-action-adjusted prices
T38.04	Point-in-time data and survivorship bias
T38.05	Spreadsheet engineering
T38.06	SQL, Python, and R for finance
T38.07	APIs and financial data pipelines
T38.08	Econometrics and financial forecasting
T38.09	Factor research and backtesting
T38.10	Look-ahead bias, overfitting, and data snooping
T38.11	Transaction costs and realistic strategy testing
T38.12	Monte Carlo simulation
T38.13	Numerical pricing and optimisation
T38.14	Algorithmic execution and systematic investing
T38.15	Machine learning for credit, fraud, and forecasting
T38.16	Natural-language processing for financial disclosures
T38.17	AI-assisted research and financial modelling
T38.18	Reproducibility, model governance, and human verification

A QUESTION TO OPEN THE AREA

Could information unavailable at the time be making this result look better?

FOLLOW THESE CONNECTIONS

T16.16 > T38.18

Reproducibility and checks support model auditability.

T25.06 > T38.12

Monte Carlo methods generate model scenarios for risk measurement.

T34.05 > T38.11

Trading costs and market impact must be reflected in realistic strategy testing.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Industry-specific finance

22 original topic units. Adapt statement interpretation, operating metrics and valuation to the industry.

TOPICS AND STABLE IDS

T39.01	Banks
T39.02	NBFCs and housing finance
T39.03	Insurance
T39.04	Asset management and brokerage
T39.05	Financial exchanges and market infrastructure
T39.06	Software, SaaS, and IT services
T39.07	E-commerce and marketplaces
T39.08	Consumer goods and retail
T39.09	Automotive and electric vehicles
T39.10	Manufacturing and capital goods
T39.11	Metals, mining, and commodities
T39.12	Oil, gas, and petrochemicals
T39.13	Power, utilities, and renewable energy
T39.14	Telecom and digital infrastructure
T39.15	Pharmaceuticals, biotechnology, and healthcare
T39.16	Real estate and construction
T39.17	Airports, airlines, ports, and logistics
T39.18	Hotels, travel, and restaurants
T39.19	Agriculture and agribusiness
T39.20	Media, entertainment, and sports
T39.21	Education and professional services
T39.22	Government enterprises and regulated businesses

A QUESTION TO OPEN THE AREA

Which metric or valuation convention changes in this industry?

FOLLOW THESE CONNECTIONS

T39.01 > T24.02

Bank analysis uses sector-specific statements and banking economics.

T39.03 > T29.11

Insurance analysis uses insurer-specific profitability and liability measures.

T39.04 > T23.04

Asset-management economics depend on assets, fees, flows and costs.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Applied cases, financial history, and integrated practice

18 original topic units. Apply multiple concepts to one decision and explain the result using evidence.

TOPICS AND STABLE IDS

T40.01	Reading a real annual report from beginning to end
T40.02	Tracing one transaction through all three statements
T40.03	Diagnosing a profitable company with a cash shortage
T40.04	Assessing whether growth creates value
T40.05	Building a three-statement forecast
T40.06	Valuing a listed company
T40.07	Writing an equity research report
T40.08	Underwriting a corporate loan
T40.09	Evaluating a capital expenditure proposal
T40.10	Comparing debt versus equity financing
T40.11	Analysing an acquisition
T40.12	Building a startup fundraising and dilution model
T40.13	Evaluating a property investment
T40.14	Designing and evaluating a portfolio
T40.15	Stress-testing a bank or business
T40.16	Investigating an accounting failure
T40.17	Studying financial crises, bubbles, and corporate collapses
T40.18	Comparing an investment thesis with the outcome

A QUESTION TO OPEN THE AREA

Can you explain the decision by connecting the statements, risks and valuation?

FOLLOW THESE CONNECTIONS

T19.12 > T40.18

Investment theses should be compared with subsequent evidence and outcomes.

T35.06 > T40.18

Confirmation bias can distort review of an investment thesis.

T40.01 > T04.16

Read disclosures and connect the accounting policies to the statements.

GO DEEPER

Use the master workbook to see every incoming and outgoing link, preparation topic, companion and authoring prompt.

Content structure for the eventual interactive guide

Concept	Stable ID, title, aliases, short meaning, deeper explanation, scope and assumptions.
Relationship	From ID, labelled relationship, To ID, explanation, conditions and mapping provenance.
Worked example	Transaction or decision, inputs, calculations, statement effects and linked concepts.
Source evidence	Document or URL, date, page or note, units, framework, jurisdiction and consolidation scope.
Question	Prompt, response options, correct explanation, hints and misconception links.
Journey	Driving question, shared concept sequence, optional branches and return context.
Learning evidence	Separate visits, practice results and demonstrated understanding. Do not equate clicks with mastery.

Authoring boundary: all original topics are mapped. The map does not claim that definitions, current tax rules, real-company datasets or validated quiz items are already complete.

Sources, coverage and interpretation

S01 Original finance inventory and interaction brief

All 40 areas and 678 original grouped topic units; preserved in the Inventory worksheet. The mapping itself is authored curriculum design.

S02 CFA Institute - Investment Foundations curriculum

Reference for breadth across markets, instruments, statements, analysis and portfolio management; not a claim that every listed topic is a CFA lesson.

<https://www.cfainstitute.org/programs/investment-foundations-certificate>

S03 CFA Institute - Candidate Body of Knowledge

Reference for investment analysis, valuation, portfolio and wealth-management coverage.

<https://www.cfainstitute.org/programs/cfa-program/candidate-resources/cbok>

S04 IFRS Foundation - IAS 7

Reference for operating, investing and financing classification. Detailed lessons must specify applicable framework and effective period.

<https://www.ifrs.org/issued-standards/list-of-standards/ias-7-statement-of-cash-flows.html/>

S05 Aswath Damodaran, NYU Stern - Approaches to Valuation

Reference for matching FCFF with cost of capital and FCFE with cost of equity.

https://pages.stern.nyu.edu/adamodar/New_Home_Page/lectures/approach.html

S06 Nielsen Norman Group - Progressive Disclosure

Reference for staged detail in the interaction architecture.

<https://www.nngroup.com/articles/progressive-disclosure/>

S07 Roediger and Karpicke - Test-Enhanced Learning

Reference for retrieval-practice questions. Specific questions and journeys in this map are proposed authoring prompts, not validated assessments.

<https://www.psychologicalscience.org/journals/psychological-science/j.1467-9280.2006.01693.x/>